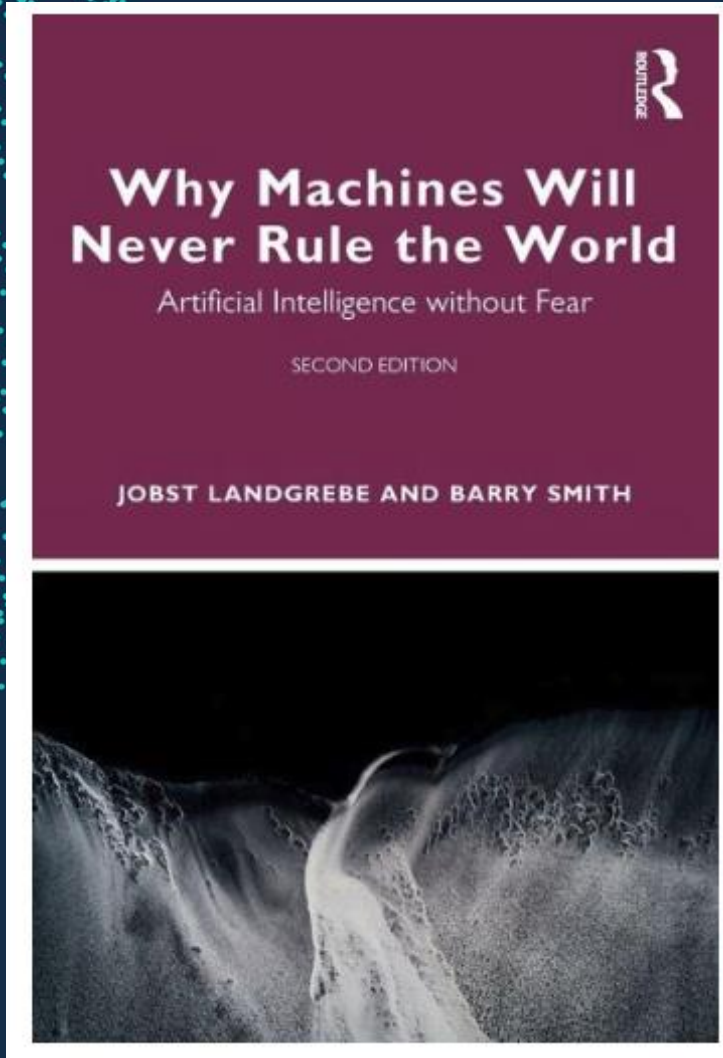


Why machines cannot write ontologies



THE WORLD'S
ONTOLOGY ECOSYSTEM

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Can there be any AI? To understand the question we need to understand human intelligence first

Practical (animal) Intelligence

This form of intelligence is *realised in humans and some higher animals*. It is the ability to adapt to new situations (Stern 1921) with the following properties.

The **adaptation**

- is **sudden** and is not something that is primed by prior experiences,
- is **meaningful**, which means that it is appropriate to the situation,
- is **untrained**, and thus is not a product of repeated attempts at trial and error,
- is **novel** from the perspective of the acting organism

Mental Intelligence

This form of intelligence is *specific to humans*. It enables:

- Temporary **independence** from immediate organic **necessities**⁽¹⁾
- View of sensual input as a matter of **objects** (animals do not have objects)
- Self-distance and distance from environment, self-consciousness, world as object (**Weltoffenheit**)
- **Personality ("personness")**, which we each perceive via our own constant stream of mental acts

Note that the "singularity" would require mental intelligence.

1) Temporary by duration of maximum length of independence interval: breathing - drinking / excreting - sleeping – eating – reproduction; no independence of immediate reflexes such as blinking, coughing and sneezing

Any AI system is made of mathematics and therefore based on regularities

Explicit (deterministic) and implicit (stochastic) AI models

$$\rho \frac{D\vec{v}}{Dt} = \rho \left(\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right) = -\nabla p + \mu \Delta \vec{v} + (\lambda + \mu) \nabla (\nabla \cdot \vec{v}) + \vec{f}.$$

Classical explicit model (partial differential equation): Navier-Stoke's momentum equation. All variables can be determined [partly analytically] exactly. **No error term.**

$$y_t = x_{t1}\beta_1 + x_{t2}\beta_2 + \dots + x_{tK}\beta_K + \varepsilon_t$$

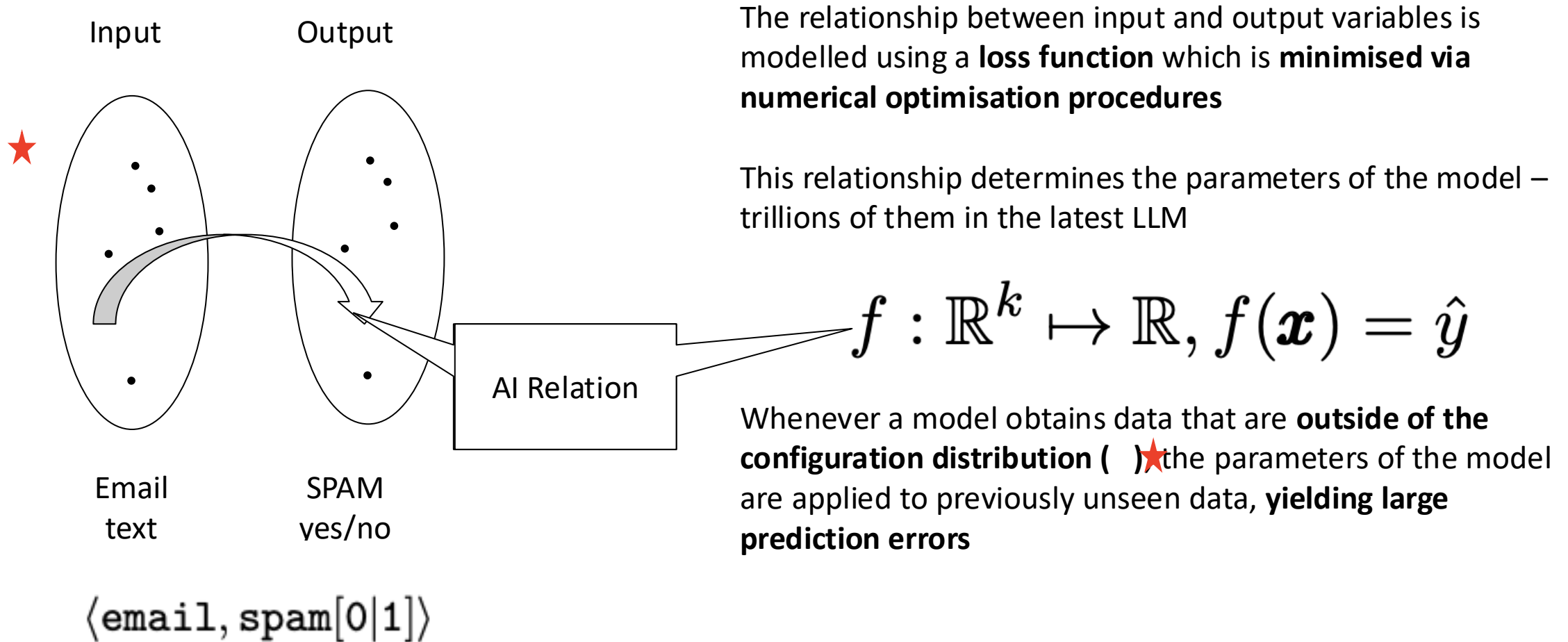
Classical implicit model: Multiple linear regression (~1800, Bošković, Legendre, Gauss). The parameters β are estimated from data **with an error ε** (stochastic model). dNN and LLM are merely very complicated regression equations.

Both methods require **regularities** and can only model or estimate regular relationships.

As we have seen, however, intelligence is characterized by irregularity, which cannot be mapped by the above methods. In addition, intelligence is a dominant characteristic of complex systems (see below).

Today, AI is mostly seen as statistical learning for automation and pattern identification purposes

Statistical (“machine”) learning



Stochastic models are very useful but scope-restricted and error-prone

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Stochastic AI problems

1. Optimisation of a problem-specific loss function, not generalisable
2. Close dependence on the selected training sample and its specific annotation
3. Failure with heterogeneous annotations for identical inputs
4. Failure with sparsely populated sample space (in stark contrast to animals and humans) - no real intelligence that can do just that
5. There is no guarantee that the result will be corrected as soon as an error is detected.
6. Algorithm understands nothing, cannot predict its own failure
7. Cannot model far-reaching relationships
8. dNN cannot model the semantics of mental categories (objectification of reality through universals), which is why they fail when interpreting images or language.
9. Insufficient accuracy for critical situations (e.g. driving a car)
10. Increasing and manipulating the training procedure leads to false generalisations

“Pope”



AI-generated



*clocks always show
10 past 10 or 10 to 2*

“First US President”



“Waffen SS soldier”



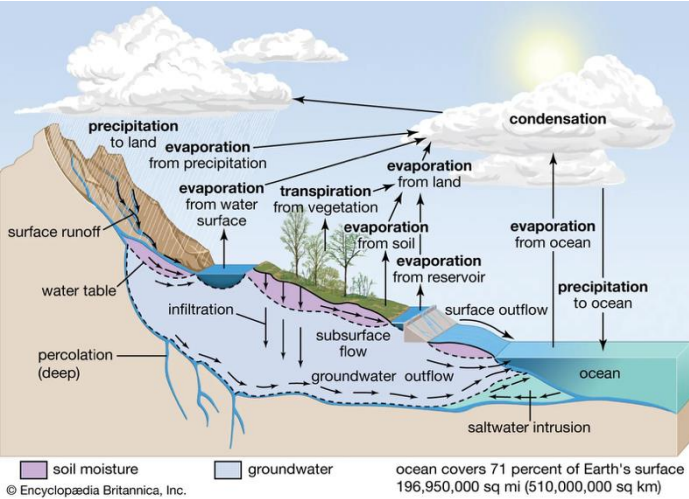
The root cause of mathematical model failure in AI is a lack of regularity in complex systems

Complex system properties – animate and inanimate		Consequence for mathematical modelling
1	Change and evolutionary character	Coordinate system may change any time
2	Element-dependent interactions	No classical laws - which need uniform and isotropic mono-interactions
3	Force overlay	
4	Non-ergodic phase spaces	○ No regularity that can be modelled as a law of physics ○ No representative samples can be taken to train implicit models
5	Drivenness	Behaviour cannot be modelled (simple example: turbulence)
6	Context-dependence	No classical laws which need context-independence
7	Chaotic behaviour	No satisfying modelling due to measurement dependence

Examples



Animate: Fly



Inanimate: Water cycle

Intelligence is a dominant property of complex living systems, as human language is

Dominant properties of complex systems

- **Complex systems** also have many properties that lead to regular patterns: In humans, it is the vital processes: breathing, heartbeat, sleep, drinking, eating, digestion, sexuality. However, these can be temporarily interrupted by the erratic character of higher functions, even breathing. **Regularity**, however, **enables their mathematical modeling**, as the perfect ECG models of the healthy and diseased heart show
- Dominant characteristics of complex systems are those that **distinguish** their bearer from other systems and **have no patterns**: In humans, these are consciousness, self-awareness, feelings, decision-making, intelligence, language.
- In the case of the sun, its dominant property as a nuclear fusion emitter as opposed to its property as a gravity mass that it shares with its satellites, which is highly regular.
- **Dominant properties of complex systems can never be mathematically mapped due to the limitation of mathematics to regular patterns.**

To write or expand an ontology, one needs world and context understanding

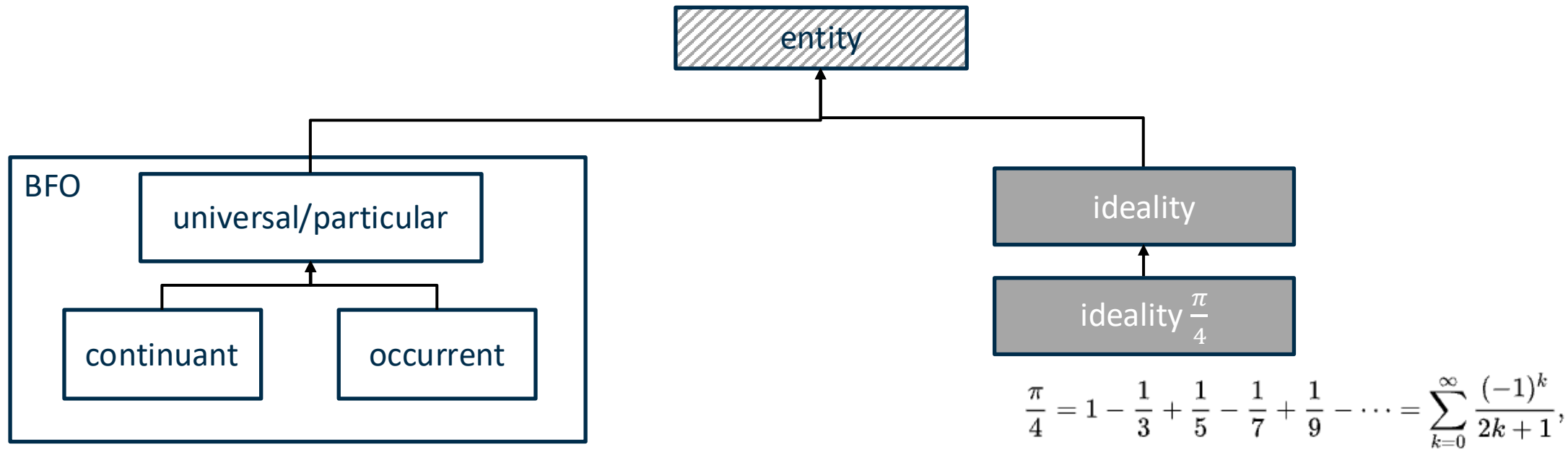
Abilities required to engineer ontologies

- Ontologies of science and engineering, but also of law or sociology, depend on our scientific understanding of the world
- Ontologies of our every-day world are seldomly formulated (such as the [W3Cs pizza ontology](#)), but they required common sense world
- Understanding of the perspective of the ontology one is writing
- Understanding of ontology authoring techniques (ontology engineering rules)
- Understanding of the scientific subject matter
 - Understanding structures of knowledge domains
 - Understanding of the evidence that supports scientific propositions
 - Relationships between scientific propositions
 - Understanding of the contentious and disputed aspects of scientific domains
- Ability to define entities in Aristotelic manner (genus species definitions)
- Ability to create new structures that are not present thus far (creativity)

Machines are not capable of this. They can merely reproduce existing patterns. Examples on the following slides.

BFO is an ontology of realism – how can one add unreal entities such as $\frac{\pi}{4}$?

Example 1: deciding on top level entities



- There is only one example in the literature to extend BFO with idealities (Landgrebe and Smith: <https://arxiv.org/abs/2305.01560>, <https://arxiv.org/abs/2606.00805>), which is a new pattern
- LLM can only work with regular patterns
- An LLM neither knows nor understands the limitations of BFO. It does not know what idealities are. It cannot expand it in any meaningful way.

Machines cannot properly avoid multiple inheritance since they do not understand the common sense or scientific definitions of entities

Example 2: multiple inheritance issue

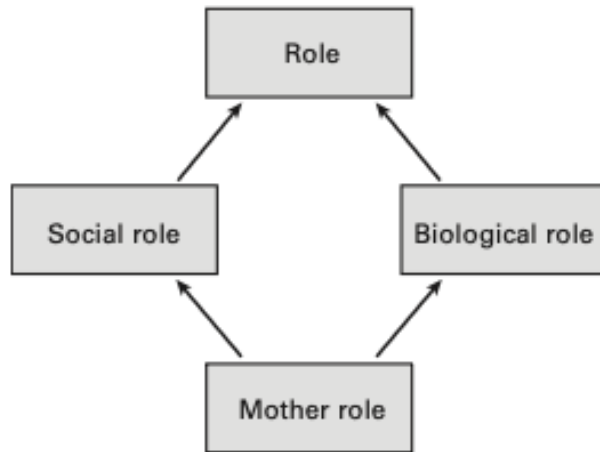


Figure 4.6
The mother role as a putative example of multiple inheritance

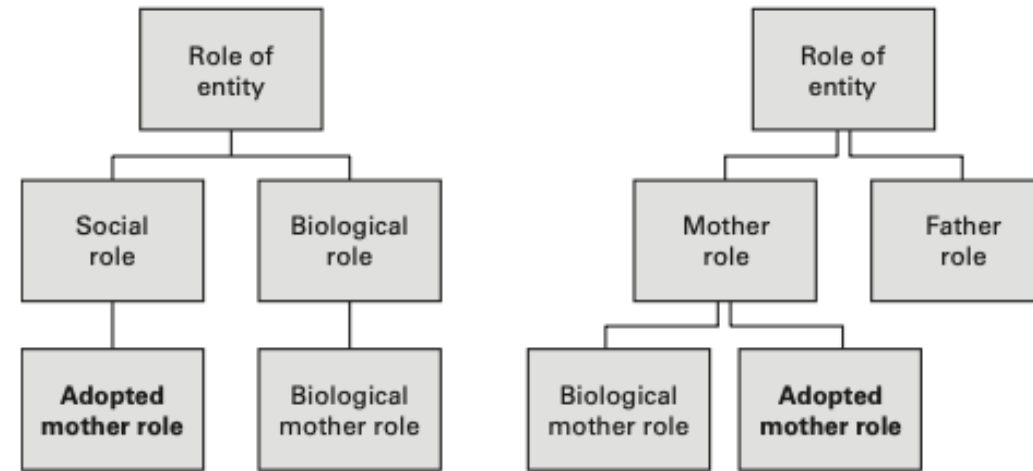


Figure 4.7
Mother classified without multiple inheritance

- While existing solutions to the multiple inheritance problem can be reproduced by LLM (with errors and hallucinations), new problems of this kind cannot be solved
- LLM are incapable of understanding the domain to an extent that allows a solution of the above type in a new situation since they cannot abstract and reason

To create an ontology, one needs to understand and apply the rules of ontology design, which machines cannot do

Example 3: ontology design details

- Include in the terminology terms of core entities in the domain to be represented.
- Strive to ensure maximal consensus with the terminological usage of scientists in the relevant discipline.
- Identify areas of disciplinary overlap where terminological usage is not consistent. Look for and keep track of synonyms.
- Ensure univocity of terms.
- Ensure univocity of relational expressions.
- Avoid mass nouns.
- Distinguish the general from the particular.
- Provide all nonroot terms with definitions.
- Use Aristotelian definitions.
- Use essential features in defining terms.
- Start with the most general terms in your domain.
- Avoid circularity in defining terms.
- ...

Machines cannot understand and apply such rules since they have no “AGI” (general artificial intelligence).

However, with ILP, one can automatically expand propositional ontologies consisting of logical clauses under restricted conditions based on input clauses

Inductive Logic Programming

Inputs

Background Knowledge (B): Pre-existing facts and rules about the domain.

Example: `parent(ann, bob)` and `female(ann)` .

Positive Examples (E^+): Specific instances that should be true.

Example: `mother(ann, bob)`

Negative Examples (E^-): Specific instances that must *not* be true.

Example: `not(mother(bob, ann))` .

Goals: The ILP system searches a space of possible logical clauses to construct a hypothesis program H such that:

Completeness: The hypothesis, combined with background knowledge, entails all positive examples:

$$B \cup H \models E^+$$

Consistency: The hypothesis does not entail any negative examples:

$$B \cup H \not\models E^-$$

Output:

In this family relationship example, the ILP system induces the general rule:

$$\text{mother}(X, Y) \leftarrow \text{parent}(X, Y) \wedge \text{female}(X)$$

An ILP system can generate ontological clauses under direct or indirect human supervision

Example generated with gemini.google.com, further reading: Nienhuys-Cheng, de Wolf: Foundations of Inductive Logic Programming (Springer, 1997); De Raedt: Logical and Relational Learning (2008)

Thanks for your attention

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Sella group, Dolomites. Winter 2026